NNN-T-001476A(GSA-FSS)
January 30, 1976
SUPERSEDING
Int. Fed. Spec. NNN-T-001476
September 8, 1970

INTERIM FEDERAL SPECIFICATION

TEST TUBE (GLASS, DISPOSABLE)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers disposable glass culture test that are lipless and suitable for one-time use in clinical laboratory procedures.

1.2 Classification.

1.2.1 Types. Test tubes shall be of the following types, as specified (see 6.1).

Type I - Borosilicate glass.
Type II - Soda-lime glass.

1.2.2 Sizes. Test tubes shall be of the following sizes as specified (see 6.1):

Size 1 - 6 x 50 millimeters (mm).
Size 2 - 10 x 75 mm.
Size 3 - 12 x 75 mm.
Size 4 - 13 x 100 mm.
Size 5 - 15 x 85 mm.
Size 6 - 15 x 125 mm.
Size 7 - 16 x 100 mm.
Size 8 - 16 x 125 mm.
Size 9 - 16 x 150 mm.
Size 10 - 18 x 150 mm.
Size 11 - 20 x 125 mm.
Size 12 - 20 x 150 mm.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

DD-G-541 - Glass (Laboratory).
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 626/GEN - Glassware; Methods for Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C 148 - Polariscopic Examination of Glass Containers.
C 149 - Thermal Shock Test on Glass Containers.
C 162 - Definition of Terms Relating to Glass and Glass Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I. Type I test tubes shall be of borosilicate glass conforming to type I of DD-G-541.

3.1.2 Type II. Type II test tubes shall be of soda-lime glass conforming to DD-G-541. NOTE: Type I, class B of DD-G-541 may be substituted at suppliers option.

3.2 Design. Test tubes shall be cylindrical with a closed, rounded bottom and open end. The open end shall be without lip, head, or flare. The inside and outside diameter of the open end shall be approximately the same as the remainder of the tube.

3.3 Annealing. Ring strain shall be not greater than that of temper no. 1, when tested as specified in 4.4.1. Any longitudinal strain shall be faint and highly diffused.

3.4 Thermal shock. Types I and II test tubes shall not crack when tested as specified in 4.4.2.1 and 4.4.2.2, respectively.

3.5 Resistance of tubes to centrifugal force. Resistance of test tubes to centrifugal force shall be such that no breakage results when the tubes are tested as specified in 4.4.3.

3.6 Dimensions. The limiting dimensions and tolerances shall be as shown in table I.

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TABLE I. Dimensional requirements

Size	Outside diameter millimeter (mm) 1/	Length (mm) 2/	Wall thickness (mm) 3/ Type I	Type II
1	6	50	0.50	0.50
2	10	75	0.60	0.60
3	12	75	0.65	0.65
4	13	100	0.70	0.70
5	15	85	0.75	0.75
6	15	125	0.80	0.75
7	16	100	0.80	0.80
8	16	125	0.80	0.80
9	16	150	0.80	0.80
10	18	150	0.85	0.85
11	20	125	0.95	0.95
12	20	150	0.95	0.95

1/ Tolerance ± 0.50 mm.
2/ Tolerance ± 2.5 mm.
3/ Tolerance ± .05 mm.

3.7 Finish. All surfaces shall be smooth and all edges rounded.

3.8 Terminology. All glass terms will be defined by ASTM C 162.

3.9 Workmanship. The test tubes shall be symmetrical in shape, of uniform thickness, and free from striae, streaks, chips, cracks, bubbles, or other defects which detract from their appearance or impair serviceability. The test tubes shall be dry, clean, and ready for use free from grease, dirt, stains, oil and/or oily-type substances.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificate of analysis. Certified test data, in lieu of contractor performing tests, will be acceptable as evidence of compliance with requirements of 3.1.

4.2 Sampling.

4.2.1 For examination. The sample unit for examination purposes shall be one test tube. The sampling shall be in accordance with MIL-STD-105.

TABLE II. Sampling for examination

Defect	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. The sample unit for testing purposes shall be one test tube. The testing shall be in accordance with MIL-STD-105.

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TABLE III. Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL
Annealing	3.3	4.4.1	S-2	*
Thermal shock	3.4	4.4.2	S-2	*
Resistance to centrifugal force	3.5	4.4.3	S-2	*

* Acceptance number shall be zero.

4.3 Examination. The test tubes shall be examined to determine compliance with all the requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects.

TABLE IV. Categories and defects

Categories	Defects
Critical	Not defined.
Major:	
101	Bottom of test tube not rounded.
102	Open end of test tube not lipless.
103	Tube not free of cracks and chips.
104	Tube not free of hairline.
105	Tube not free of sharp or rough edges.
106	Tube not free of blisters or checks.
107	Tube not free of dirt, cloudiness, or foreign matter.
108	Tube not clean and clear.
Minor:	
201	Tube not free of stones and striae.
202	Tube not free of bubbles.

Examination shall not be restricted to defects listed herein. Definition of glass defects shall be in accordance with ASTM C 162.

4.3.2 Dimensional examination. The test tubes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

TABLE V. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing.

4.3.4 Examination for closure of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure shall be in accordance with appendix of that specification.

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4.4 Tests.

4.4.1 Annealing. Test tubes shall be tested for annealing according to ASTM C 148.

4.4.2 Thermal shock.

4.4.2.1 Type I. Type I test tubes shall be tested for thermal shock in accordance with Fed. Test Method Std. No. 626, method 401.

4.4.2.2 Type II. Type II test tubes shall be tested for thermal shock in accordance with ASTM C 149.

4.4.3 Resistance to centrifugal force. Test tubes of the same size shall be filled with an equal amount of distilled water and suspended directly across from each other in a centrifuge. The centrifuge shall have an angle head of 40° to 50° from the vertical, and a radius of 1-1/2 to 1-3/4 inches from the center of the tube shield. The centrifuge shall be accelerated gradually to a speed of 1500 r.p.m., for sizes 10, 11, and 12 and 3000 r.p.m., for sizes 1 thru 9. This acceleration shall continue for a period of 5 minutes. After the period of acceleration, the centrifuge shall be allowed to decelerate without the use of a brake.

4.5 Acceptance/rejection criteria. If any test tube fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Test tubes shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level B or C as specified (see 6.1).

5.2.1 Level B. Five hundred (500) or one thousand (1,000) tubes of one type and size as specified (see 6.1) packaged in accordance with 5.1 shall be packed into a box conforming to PPP-B-636, class domestic. The test tubes shall be protected by cells, partitions, other devices or shrink wrapped in groups or in total to prevent shifting or damage within the box. The box shall be closed in accordance with the appendix of PPP-B-636.

5.2.2 Level C. Test tubes in quantities as specified (see 6.1), shall be packed in a container complying with the Uniform Freight Classification or National Motor Freight Classification as applicable. The containers shall be packed to assure carrier acceptance and safe arrival at destination.

5.3 Marking. In addition to any special marking required in the procurement document, marking of the shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2).
(c) Level of packing required (see 5.2 and 5.2.1).
(d) Quantities required (see 5.2.2).

GSA DC 76-724

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